

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43
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COURSE OUTCOME COVERED

CO-2: Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool: Short Answer Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I K.GOUTHAM (192311169) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *k.goutham..*

RUBRICS FOR CALCULATION:



Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with	Incorrect or irrelevant answer



				noticeable errors	
Application of Knowledge	20	Correctly applies concepts to solve the question/ problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

CO2 - AT3Question 1

Responsive Web Design (RWD) is a web design approach that makes a website automatically adjust its layout, images, and content according to different screen sizes such as mobile phones, tablets, and desktops.

Importance:

- Provides better user experience on all devices
- Reduces the need for separate mobile and desktop website
- Improves website accessibility.
- Enhances SEO (Search Engine Optimization).
- Increases user satisfaction and engagement.

KEY CSS Techniques

1. Fluid Layouts - Use percentage (%) instead of fixed pixels
2. Flexible Images - Use max-width: 100%.
3. Media Queries - Apply different styles for different screen sizes.
4. Relative Units - Use em, rem, %, and vw.

Flexbox/Grid for Adaptive layout

- Flexbox is used for one-dimensional layouts (rows or columns)
- CSS Grid is used for two-dimensional layouts (rows and columns).

HTML

```
<div class = "container">  
  <div> Home </div>  
  <div> About </div>  
  <div> Contact </div>  
</div>
```

CSS

• container

display: flex;

flex-wrap: wrap; y

• container div {

flex: 1;

padding: 20px;

border: 1px solid black; y

Improvement:

- Use Responsive images
- Optimise loading speed
- " provide touch-friendly buttons
- Use readable fonts
- Test on multiple devices.

Question 2:

client-side Validation.

client-side validation check user input in the browser before sending data to the server.

Purpose:

- Prevents invalid data submission
- Reduces server load.
- Gives immediate feedback
- Improves user experience.

Javascript Event handling

Javascript uses events like:

- submit • click • keyup • change

to validate user inputs.

Role of Regular Expressions.

Regular Expressions (Regex) are patterns used to validate input forms.

Example:

- Email
- phone Number
- Password.

Example

HTML

```
<form onSubmit = "return validate()">
```

Email:

```
<input type = "text" id = "email" >
```

```
<input type = "submit" >
```

```
</form>
```

```
<function validate() {
```

```
let email = document.getElementById("email");
```

Improvements:

- Display error messages beside input fields.
- Validate passwords with strong Rules.
- Use HTML5 validation attributes.
- perform server-side validation for security.
- Sanitize user inputs.

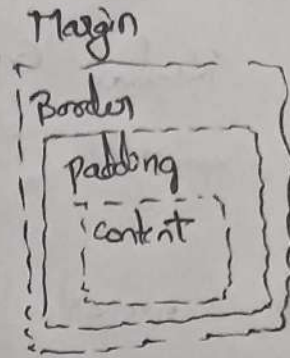
Question 3 (5 Marks)

CSS Box Model

The CSS Box Model describes how elements occupy space on a webpage.

Components:

1. Content - Actual text or Image
2. Padding - Space around content
3. Border - Surrounds padding
4. Margin - Space outside the border



CSS Positioning Techniques

- static - Default position.
- Relative - Moves relative to its normal position
- Absolute - Positioned relative to the nearest position ancestor
- Fixed - Fixed on the screen
- Sticky - Switches between relative and fixed.

Causes of Layout Breaking

- Fixed width elements
- Large images
- Incorrect positioning
- Overflow issues
- Wrong media queries

Media Queries

@Media (max-width: 768px) {

• container
width: 100%;
display: block;

}

}

Best Practices:

- Use Flexbox or Grid
- Avoid fixed widths
- Use Relative units
- Apply media queries
- Test on different devices and browsers

Question 4's

Document Object Model (DOM)

The Document Object Model (DOM) is a tree-like structure that represents HTML elements as objects. JavaScript can access and modify these objects dynamically.

ROLE OF DOM

- change webpage content.
- Modify styles.
- Add or remove elements.
- Handle User Interactions.
- Update page without reloading.

DOM Manipulation Example:

```
<p id="msg">welcome </p>
```

```
<button onclick="changeText()">click </button>
```

```
<script>
```

```
function changeText() {
```

```
    document.getElementById("msg").innerHTML = "Welcome to JavaScript";
```

```
</script>
```

When the button is clicked, JavaScript changes the paragraph text without reloading the page.

Improvements :

- Minimise unnecessary DOM updates
- Use event delegation for better performance
- Update only the required elements.
- Use efficient Javascript code.
- Optimise animations for smoother user experience.